

THE SOLUTION - 4 CORE SUBSYSTEM
MOATS

SWARM
ARCHITECTURE

AI Swarm Drones Find Survivors **3-4x** **Faster** Than Traditional SAR

10-18
MIN/MI²

78-85%
DETECTION

12% FALSE
POSITIVES

\$12K
SORTIE



50Hz
OFFBOARD

SUBSYSTEM A

**SUTRA-
FSD GNC &
3D ORCA**

**SOLVES
PROBLEM VOID:**
Canopy GPS
Multi-Path
Drift &
Crashes

**SOLUTIONAL
MOAT:**

Decentralized
3D
velocity
obstacle
collision
avoidance
+ PX4
MicroXRCE
50Hz
offboard
setpoint
streaming.

0.05s
Response
Latency

0
Tower
Dependency



-5dB
RESILIENT

SUBSYSTEM B

**Deep
JSCC
Neural
Video**

**SOLVES
PROBLEM VOID:**
Mountain
Ravine RF
Ridge
Blackout

**SOLUTIONAL
MOAT:**

Differentiable
joint
source-
channel
coding
yielding
zero
digital
cliff and
graceful
video
degradation
down to
-5dB.



3.59cm
ACCURACY

SUBSYSTEM C

**3D DEM AI
Geolocation**

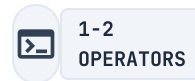
**SOLVES
PROBLEM VOID:**
35% False
Alarms & Flat
GPS Drift

**SOLUTIONAL
MOAT:**

Jetson
Orin
YOLOv8-
Pose with
real-time
WGS84
elevation
raycasting,
eliminating
15-30m
calculation
errors.

<15ms
Inference
Time

3.59cm
Geo
Error



1-2
OPERATORS

SUBSYSTEM D

**Pegasus
3D
WebGPU
GCS**

**SOLVES
PROBLEM VOID:**
15-25 Ground
Crew
Cognitive
Overload

**SOLUTIONAL
MOAT:**

Offline
tactical
3D
digital
twin +
ATAK
plugin,
empowering
1-2
operators
to
orchestrate
full

-5dB

SNR

Threshold

0

Cliff

Degradation

drone

swarms.

1-2

0ns

100%

Offline